

Electro-gastrogram (EGM) EGM represents the record to potentials associated with muscle activity in the gastro-intestinal tract. It is a useful tool for the clinicians to determine the lesions/injuries of the various gastro-intestinal muscles.

17.7 ENVIRONMENTAL AIR POLLUTION PARAMETERS

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Environmental air pollution can be defined as the addition to our atmosphere of any material(s) having a harmful effect on our lives. Typical gaseous pollutants include gases such as carbon monoxide (CO), carbon dioxide (CO₂); compounds of sulphur like sulphur dioxide (SO₂); hydrogen sulphide (H₂S); oxides of nitrogen like nitrous oxide (N₂O), nitric oxide (NO), nitrogen dioxide (NO₂); ozone (O₃); hydrocarbons, organic para-oxides, aldehydes, organic acids and other partially oxidised organic compounds including nitrogen containing esters such as peroxyacylnitrates (PANs). In addition, the other pollutants are vapours of water and other solvents present in smoke; aerosols such as inorganic sulphates, nitrates, chloride and ammonium salts and particulates which are small particles usually of solids such as atmospheric dust, coal dust, fly ash, insecticide dust, pollen, metallic foundry dust, milled flour, etc.

Air pollutants can be classified as follows:

1. *A primary pollutant* It is an emission of high concentration which originates from the source. It is usually lethal in nature.
2. *A secondary pollutant* It is an emission of considerably lower concentration which is formed through the reaction of primary pollutants. This reaction can occur at the emission point or at a remote location.

Air pollution is mainly caused by the exhaust emissions of various process industries, combustion type vehicles, thermal power stations, domestic heating, forest and agricultural fires, etc. The various components of air pollution can have quite a harmful effect on human health. For example, CO is known to cause heart disease. SO₂, NO, NO₂ and the particulates are irritants to breathing and create discomfort to the eyes. Further, inhalation of O₃ in large quantities may lead to a severe headache.

The quantities of pollutants can be expressed either on a volumetric or mass basis. The commonly used volumetric unit for gaseous pollutants is parts per million which is abbreviated as ppm and is defined as:

$$1 \text{ ppm} = \frac{1 \text{ volume of gaseous pollutant}}{10^6 \text{ volumes of air (containing pollutants)}} \\ = 0.0001 \text{ per cent by volume} \quad (17.20)$$

Alternatively, the amount of pollutant in the air can be expressed in mass units, i.e. in the form of $\mu\text{g}/\text{m}^3$. Further, it is possible to convert the quantities in ppm to $\mu\text{g}/\text{m}^3$ knowing the molecular weight of the particular gaseous pollutant and assuming the ideal gas equation to hold good at the ambient temperature and pressure conditions.

The following are the commonly used devices employed in the environmental air pollution studies.

17.8 ENVIRONMENT POLLUTION MEASUREMENT TECHNIQUES AND DEVICES

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17.8.1 Orsat Apparatus for Exhaust Gas Analysis

The analysis of boiler flue gases or the emissions of power generating combustion type engines is important not only to the air pollution studies but also to optimising the combustion phenomenon for obtaining the most efficient burning rates resulting in better energy utilisation.

A simple and commonly used laboratory device, namely Orsat apparatus shown in Fig. 17.27, is employed to analyse the products of combustion, i.e., to determine the volume fraction of a mixture of CO_2 , O_2 , CO and N_2 . This is carried out by selective absorption of individual components by the reagents and the consequent reduction in volume represents the volume fraction of the sample gas. A sample of the products of combustion is taken into the measuring burette using a hand-operated or hydraulic aspirator or a small electric pump. The sampling inlet valve is shut off and the sample is forced into the first reagent pipette containing caustic potash (KOH) solution to absorb only CO_2 . The sample is then brought back to the measuring burette and the decrease in volume from the original volume is recorded to determine the percentage of CO_2 . This procedure is successively repeated with the other two reagent pipettes containing pyrogallic acid and cuprous chloride solutions for the absorption of O_2 and CO , respectively. Finally, the proportion of N_2 is determined by difference. From the volumetric analysis, the dry molecular weight of the sampled gas can be determined as follows:

$$M_D = 0.44 (\% \text{ of } \text{CO}_2) + 0.25 (\% \text{ of } \text{CO}) + 0.32 (\% \text{ of } \text{O}_2) + 0.28 (\% \text{ of } \text{N}_2) \quad (17.21)$$

where M_D = dry molecular weight of the sampled gas.

It may be noted that the reagents of the apparatus are to be replaced when exhausted. Further, this device cannot be used for continuous monitoring.

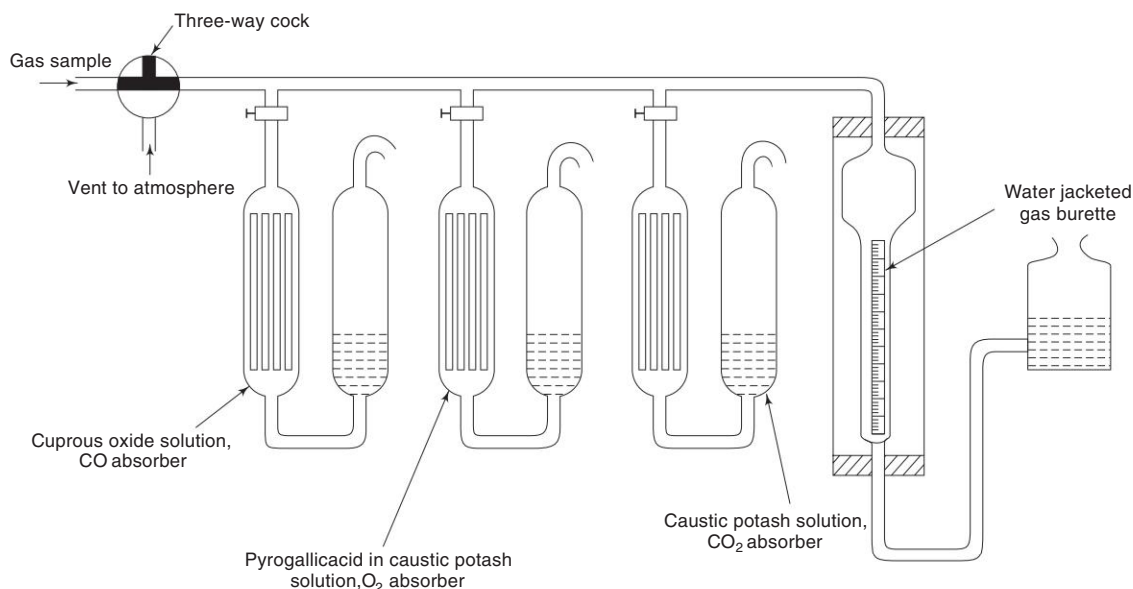


Fig. 17.27 Orsat gas analysis apparatus

17.8.2 Gas Chromatography

Gas chromatography is one of the most popular techniques of physically separating the constituents of complex mixture of gases and volatile compounds. After this the concentrations of the individual components are determined using a suitable gas chromatograph (GC) detectors. The notable feature of this technique is that it has strong separating power and even complex mixtures can be resolved into its constituents. The selective sampling of various components takes place in the gas chromatograph (GC) column which is packed with a fine mesh of adsorbing material such as activated charcoal, silica gel or

activated alumina. Alternatively, it may be filled with a suitable adsorbent liquid depending on particular component of gas to be sampled.

The test sample gas mixture for analysis in gas chromatography is usually kept small. It is typically collected in glass syringe and a small metered volume is introduced into the chromatographic column. Immediately after injection the various components of sample get selectively adsorbed by the solid/liquid adsorbents of the GC column. After this, an inert carrier gas, usually ultra pure nitrogen or argon or helium is fed at constant flow rate which moves the different constituents of the test gas mixture at different speeds and hence they get separated, as the mixture covers a long path through the column. In other words, the different components of the sample gas have different *adsorption times* or *retention times* in the GC column and are *eluted* or *released* at different rates. Consequently, the output of the GC detector termed as chromatogram, is in form of distinct peaks of constituent gases of the test gas mixture, each separated by a small time interval. The device is usually calibrated for retention times of various components of sampled gas and the concentration of each gas is determined from the separated sample pulse of GC column. Chromatographic devices are equipped with suitable closely controlled heating or cooling systems to facilitate proper separation times for different components of sample gas mixture.

A schematic diagram of the gas chromatograph is shown in Fig. 17.28. It consists of four major components viz.

- (i) High pressure carrier gas supply bottle equipped with pressure regulator and flow rate monitor.
- (ii) Sample injection unit
- (iii) *Chromatographic column* packed with *adsorbents* and enclosed in temperature controlled oven.
- (iv) GC detector unit along with associated electronic circuitry for recording and display of chromatograms.

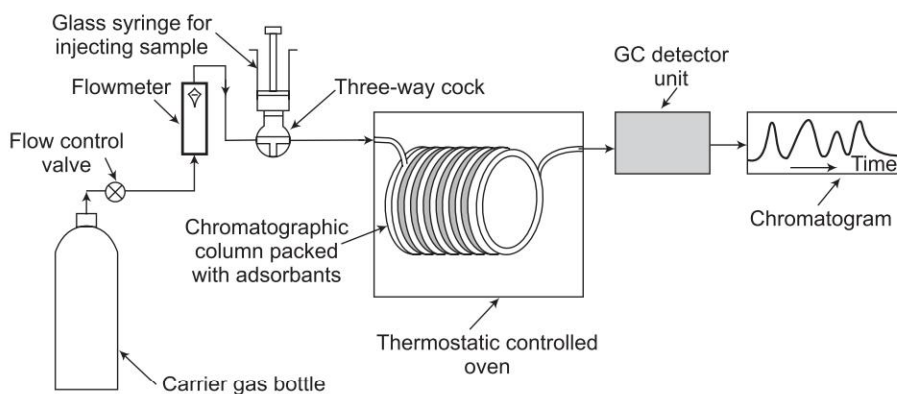


Fig. 17.28 Block diagram of Gas Chromatograph

The chromatographic packed column also termed as fractionating column is considered as heart of the system. It is in the form of a coil made of copper or stainless steel tube of 0.25 to 50 mm in diameter and 1 to 20 m in length.

The detector unit of the system generates the signal in electrical form which is proportional to the concentration the component. The display in the form of chromatogram indicates the concentration curves or peaks of the various components of the mixture. The qualitative identification of particular component is made by the retention time, which is the time elapsed from injection of the sample to its detection.

Chromatographic units are available for 100 or more constituents in the mixture with detection sensitivity of the order 0.005 ppm with accuracy limits of $\pm 2\%$. The other notable features of instruments are its requirement of small quantity of sample and results are obtained in the fixed time frame of 10 to 30 minutes. Presently portable units are also available with multicolour strip chart recorder, calibration gas, microprocessor based peak analysis and a recorder which prints out the compounds by name and concentration, based on stored library. Further, they can be operated by either battery or mains supply.

17.8.3 Gas Chromatograph Detectors

The ideal detector for gas chromatography should have the following characteristics:

- (i) adequate sensitivity
- (ii) good stability and reproducibility
- (iii) a linear response
- (iv) ability to withstand temperature ranging from room temperature to at least 400°C .
- (v) high reliability and convenience in usage and
- (vi) highly selective response towards different constituents of the sample mixture

A number of GC detectors are available using a variety of techniques for determining the concentration of constituents of test samples. Some commonly used GC detectors have been discussed in the following section.

(i) Thermal Conductivity Detector (TCD) It consists of four insulated bulbs, each of which has identical helical filament of either Teflon covered Tungsten or Gold sheathed Tungsten or solid state matched thermistor. In other words, each container which is called cell has the matched resistance element. They form a Wheatstone bridge, which is operated in full bridge configuration in which two cells are measuring cells and the other two are reference cells as shown in Fig 17.29. If the carrier gas flows through all the cells, then bridge would be balanced. As soon as, the sampled gases from the GC column, enters the measuring cells in two opposite arms of the Wheatstone bridge, the resistances of measuring cells would change. This is because the thermal conductivity of sampled gases would be different from that of the carrier gas. The unbalanced bridge output voltage is proportional to concentration of sample gas and the potentiometric type strip chart recorder connected with TCD gives the desired chromatogram of the test sample.

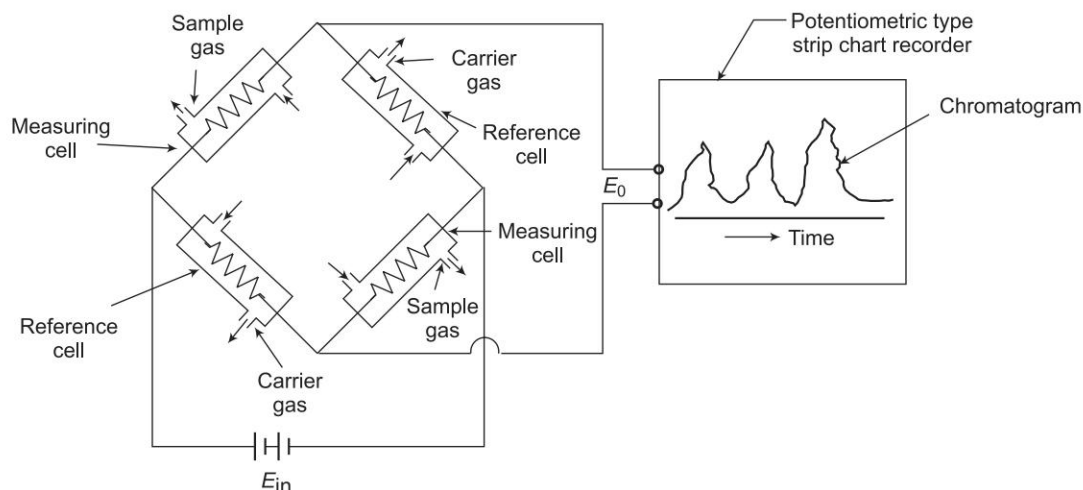


Fig. 17.29 Thermal Conductivity Detector (TCD)